Activity-1

Medical Image enhancement.

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1) Gray-level transformation:-

I recommend contrast switching to improve the visibility of darker tissues in the x-ray image.

2) why is it suitable?

Contrast stretching increases the range of gray levels in the image. It improves the contrast between darker tissues and brighter bones, making the details easier to identify.

3) what effect will it have on the image?

The image will have better contrast. Darker tissues will become easier to distinguish, and the bones and surrounding tissues will be more clearly visible.

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2) Night-Time Surveillance:-

1) Transformation:-

I recommend Log Transformation to improve the visibility of the dark CCTV image.

2) Suitable?

Log Transformation expands the low-intensity pixel values and compresses the high-intensity values. Since most pixel values are between 0 and 40,

it brightens the dark regions effectively. Negative transformation reverses the brightness values but does not specifically enhance dark intensity differences in the same way.

3) Predict the appearance of enhanced image.

The enhanced image will appear brighter. objects such as vehicles, people and other details in the dark parking lot will become more visible.

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3) Fingerprint Recognition:-

1) which gray-level transformation would you apply?

I would apply contrast stretching to improve the contrast between fingerprint ridges and valleys.

2. Explain why?

Contrast stretching expands the gray-level range, making the ridges and valleys easier to distiguish. This improves fingerprint detail visibility and supports accurate recognition.

3. what would happen if a log transformation were applied instead?

Log transformation would brighten the darker regions and compress the brighter regions. However, it might not improve the contrast between ridges and valleys as effectively as contrast stretching.